



14-763/18-763

Recitation 2: Introduction to Google Cloud and Agentic Engineering

FALL 2026

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Agenda

Part 1: Cloud Foundations & Tools

- 1. Google Cloud Essentials:** What is GCP, cloud vs. local compute, projects, APIs, and pricing calculator
- 2. GCP Coupon Redemption:** Step-by-step credit activation via Canvas & personal Gmail
- 3. Operational & Financial Governance:** Stopping VMs, disabling billing, and \$30 budget alerts
- 4. Cloud Console & Essential APIs:** UI tour, omnibar navigation, and enabling course APIs
- 5. Developer Tools & Cloud Shell:** Interactive browser terminal, local CLI, and 2026 modernizations
- 6. Google Cloud CLI (`gcloud`):** Installation, initialization, and Application Default Credentials (ADC)

Part 2: Infrastructure, Security & AI

- 1. Platform Overview - Compute:** Compute Engine VMs, machine types (E2 to GPU rigs), & Dataproc
- 2. Data & Storage Ecosystem:** Cloud Storage buckets, modern `gcloud storage`, & databases
- 3. Google Cloud IAM & Logging:** Least privilege security, service accounts, & Logs Explorer
- 4. Antigravity CLI & VS Code:** Cross-platform installation, VS Code extension, and verified slash commands
- 5. Vertex AI Browser Auth:** Interactive Google Cloud login, linking course project, & zero env vars
- 6. Recitation Summary & Checklist:** Next lecture student action items, support channels, and Q&A

What is Google Cloud?

Google Cloud Platform (GCP): A comprehensive suite of cloud computing services running on the same hyperscale infrastructure Google uses for its consumer products (Search, YouTube, Gmail).

Core Service Domains: Provides on-demand access to virtualized computing, scalable object storage, global low-latency networking, managed databases, big data analytics, and enterprise AI/ML models.

Pay-As-You-Go Economics: Rent resources dynamically by the second rather than purchasing and maintaining physical hardware servers.

Explore Available Products & Pricing Calculator:

Product Catalog: Visit <https://cloud.google.com> to explore full product specs, documentation, and architecture whitepapers.

Google Cloud Pricing Calculator: Use <https://cloud.google.com/products/calculator> to estimate exact monthly costs for compute, storage, and networking before provisioning resources!

Cloud vs. Local Laptop for GenAI Workloads

Why Local Hardware Fails for Production GenAI: Modern generative AI models, embeddings pipelines, and multi-agent loops rapidly exceed standard consumer laptop capabilities.

Local Laptop Constraints

- **Restricted Memory:** 16 GB – 32 GB RAM shared between OS, IDE, and model weights.
- **Limited Compute:** Consumer CPUs/GPUs suffer thermal throttling under sustained load.
- **Single Point of Failure:** If laptop crashes or disconnects, all running experiments terminate.
- **Environment Drift:** Inconsistent Python/CUDA versions make sharing code difficult.
- **Isolated Data:** Multi-gigabyte training sets cannot be shared easily with team members.

Google Cloud Advantages

- **Elastic Scale:** Instantly spin up dozens of vCPUs and hundreds of gigabytes of RAM.
- **GPU / TPU Accelerators:** On-demand access to NVIDIA L4, A100, H100, and Google TPU v5e.
- **Durable Persistence:** Automated snapshots, persistent disks, and background job resilience.
- **Reproducible Runtimes:** Consistent containerized environments across all course students.
- **Centralized Storage:** Shared Cloud Storage buckets for unified datasets, outputs, and embeddings.

Essentials of GCP: Projects & APIs

Unified Resource Organization: GCP provides a centralized platform where you configure billing, provision infrastructure, and enforce security policies regardless of physical data center location.

The Project: The Core Container of GCP: Every resource you create (VM, storage bucket, cluster, dataset) belongs to exactly one project.

Project Name: Human-readable display name chosen by you (e.g. `CMU GenAI Recitation 1`). Can be modified anytime.

Project ID: Globally unique, immutable string identifier used by the `gcloud` CLI, REST APIs, and Antigravity agents (e.g. `cmu-genai-f26-student12`).

Project Number: Automatically generated 12-digit numeric identifier used internally by GCP control systems.

Identity & Access Management (IAM): Controls who (users, service accounts) has what permissions on which resources across the project.

The Foundational Principle of GCP:

[INFO] Architecture Insight: GCP is an API-First Platform

Google Cloud is fundamentally a collection of secured REST and gRPC APIs. The Cloud Console is simply a visual web UI that issues calls to those exact same APIs. Everything you click can be automated with code and CLI scripts.

Official Video Guide: https://www.youtube.com/watch?v=4D3X6XI5c_Y

Getting Started with Google Cloud: Lifecycle

Standard 7-Step Cloud Engineering Workflow

- 1. Sign In:** Authenticate with your approved Google Account in Google Cloud Console.
- 2. Select or Create a Project:** Create your dedicated project or select the assigned course project.
- 3. Enable Billing & Link Credits:** Bind your educational coupon credits to ensure resources are funded without personal charges.
- 4. Enable Required APIs:** Activate the specific APIs required for your lab (Compute Engine, Dataproc, Vertex AI).
- 5. Provision Resources:** Launch virtual machines, Cloud Storage buckets, or managed Dataproc Spark clusters.
- 6. Monitor Usage & Logs:** Track CPU/RAM metrics and review real-time logs in Logs Explorer to verify correct behavior.
- 7. Stop & Clean Up Resources:** Shut down VMs and delete ephemeral clusters immediately upon lab completion to conserve credits.

Official Video Guide: <https://www.youtube.com/watch?v=GKEk1FzAN1A>

GCP Coupon Redemption: Fundamental Rules

Rule 1: Use a Personal Gmail Account ONLY When redeeming your GCP educational coupon code, you MUST use a standard personal Gmail address.

Strict Policy: Please DO NOT use your Andrew email (@andrew.cmu.edu) to redeem coupons on Google Cloud. Enterprise-managed institutional accounts cannot accept educational promotional codes.

Rule 2: Check for Free Trial Credit Conflicts Your personal Gmail account should NOT have active Google Cloud Free Trial credits (\$300 trial).

Best Practice: If you are unsure whether your Gmail account previously used free trial credits, create a brand-new Gmail account dedicated exclusively to this course.

Rule 3: One Coupon per Student Each promotional coupon code can only be redeemed once and is bound to a single billing account.

[WARNING] CRITICAL ACCOUNT POLICY: Do NOT Use Andrew Email on GCP Console

CMU Andrew Google accounts are managed Google Workspace accounts with strict institutional policies. Attempting to redeem educational coupons under your Andrew login will result in redemption failure. Always use a personal @gmail.com account for the GCP Console.

Redeeming Your GCP Coupon: Step-by-Step

Step 1: Access Canvas Link Click on the GCP Coupon Redemption Link published in the Canvas course modules.

Step 2: Enter Student Credentials Fill in your First Name, Last Name, and your official @andrew.cmu.edu email ID.

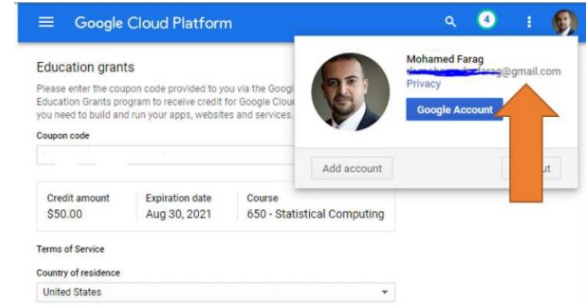
Step 3: Verify Your Email Check your Andrew email inbox for an automated verification email. Click the verification link inside.

Step 4: Receive Coupon Code You will receive a second email to your Andrew address containing your unique 16-character coupon code. Copy it.

Step 5: Navigate to Education Portal Open <https://console.cloud.google.com/education> in your browser.

Step 6: Switch Account Prior to Pasting Look at the top-right corner avatar. Ensure you are signed in with your personal Gmail. Switch accounts if necessary!

Step 7: Redeem Paste the coupon code into the text field and click Accept and Continue.



Coupon Distribution & Validating Credits

[INFO] Note on Coupon Distribution

Coupons will be distributed in the upcoming lecture. Please review this workflow carefully so that you are fully prepared to redeem your credits immediately upon distribution.

How to Validate Successful Credit Redemption

- **Navigate to Billing Section:** In the Google Cloud Console, open the Navigation Menu (top left) and click 'Billing'.
- **Inspect Credits Menu:** From the left navigation menu of the Billing dashboard, select 'Credits'.
- **Verify Amount & Expiration:** Confirm your promotional credit balance (e.g. \$50.00), credit name, and expiration date.
- **Link Billing to Your Project:** Under Billing > 'Account management', verify that your recitation project is linked to this funded billing account.

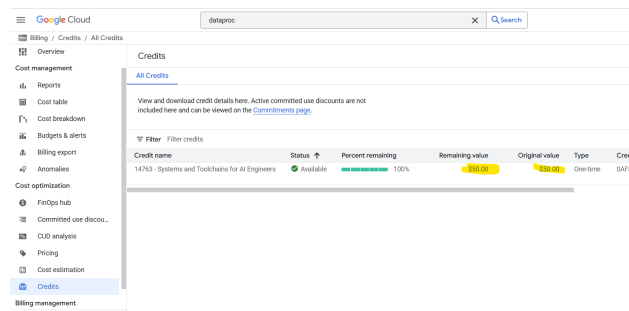
Operational Note #1: Stopping Compute & Disabling Billing

Always Stop Compute After Use: Virtual machines and Dataproc clusters consume credits every second they are in the `RUNNING` state.

Nightly Rule: Never keep your VM or Dataproc cluster running overnight! Idle instances will rapidly drain your semester credit balance.

Wait for Complete Shutdown: Before disabling billing, wait for your VM instance or cluster to completely reach the `TERMINATED` / `STOPPED` state in the console.

How to Disable Billing (End of Lab): Navigate to Billing > 'Account management'. Locate your project, click the three vertical dots (⋮) on the right, and select 'Disable billing'.



The screenshot shows the Google Cloud Credits page. The left sidebar contains navigation links for Billing, Credits, All Credits, Overview, Cost management, Reports, Cost table, Cost breakdown, Budgets & alerts, Billing export, Anomalies, Cost optimization, FinOps hub, Committed use discount, CPU analysis, Pricing, Cost estimation, Credits, and Billing management. The main content area is titled 'Credits' and shows a table of credit details.

Credit name	Status	Percent remaining	Remaining value	Original value	Type	Cred
14763 - Systems and Toolchains for AI Engineers	Available	100%	\$25.00	\$25.00	One-time	DAF

Operational Note #2: Re-Enabling Billing Safely

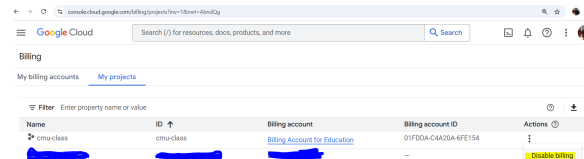
The 5-Minute Propagation Delay: When you return to work on a lab, re-link your billing account to your project under Billing > Account management.

Crucial Waiting Period: After enabling billing, wait 5 minutes before attempting to start your Dataproc cluster or VM instances.

Why? Google Cloud's internal billing authorization tokens take a few minutes to synchronize across all global API endpoints.

Verify Cluster Instances: Navigate to Compute Engine > VM Instances. Ensure all nodes transition cleanly to `RUNNING`.

Troubleshooting Failed Starts: If your VMs fail to start or remain in an error state, stop the Dataproc cluster completely and initiate a fresh start.



Operational Note #3: Running Out of Credits

Proactive Monitoring: Monitor your credit consumption regularly by checking the 'Credits' tab in the Billing section.

Never Wait for Complete Depletion: If you run out of credits completely, GCP will suspend running VMs and reject API requests, disrupting your coursework.

Emergency Credit Request Protocol: If you are nearing the end of your allocated credits and have upcoming assignments:

Contact Course Instructor for Credit Replenishment

- **Instructor Email:** Send an email directly to Prof. Mohamed Farag at farag@cmu.edu
- **Subject Line Format:** [14-763/18-763 Credit Request] Student Name - Andrew ID - Project ID
- **Required Information:** Include your current Project ID, remaining credit balance, and the specific assignment context.
- **Advance Notice Required:** Allow 24 to 48 hours delay to receive a new coupon code. Please request replenishment well before deadlines!

Setting Budgeting Alerts on Billing Accounts (~\$30)

[IMPORTANT] Set a \$30 Budget Alert to Prevent Unintended Personal Charges

Setting a budget alert ensures you receive immediate automated email notifications before resource usage exceeds expectations. Every student must configure a \$30 budget alert.

Step-by-Step Budget Alert Configuration in Google Cloud Console

- 1. Open Budgets & Alerts:** In Cloud Console, navigate to Navigation Menu > Billing > 'Budgets & alerts'.
- 2. Click '+ Create Budget':** Set the budget name to: 'Fall2026-Student-Budget'.
- 3. Define Scope:** Select your active course project and ensure all GCP services are included.
- 4. Specify Target Amount:** Under 'Amount', choose 'Specified amount' and enter exactly \$30.00.
- 5. Configure Multi-Tier Thresholds:** Set alert percentage triggers: 50% (\$15.00), 90% (\$27.00), and 100% (\$30.00) of budget.
- 6. Notification Delivery:** Check the box: 'Email alerts to billing account administrators and users' to receive alerts in your inbox.

Google Cloud Console Tour

Access Point: Navigate to <https://console.cloud.google.com> to access the central web management console.

Key Visual Navigation Components:

Navigation Menu (Hamburger Icon, Top Left): Access all GCP products organized by domain: Compute Engine, Cloud Storage, Dataproc, IAM & Admin, Billing, and Vertex AI.

Project Picker (Top Left, Beside Google Cloud logo): Always verify you have the correct course Project ID selected before creating or viewing resources!

Omnibar Search (Top Center): Search for any service, resource name, VM instance, or official documentation page with keyboard shortcut ``/``.

Cloud Shell Activation (``>_`` Icon, Top Right): Launches an instant, pre-authenticated Linux terminal inside your browser.

Notification Bell & Cloud Activity (Top Right): Displays real-time asynchronous progress of operations (VM creation, cluster provisioning, disk resizing).

Enablers for Course Infrastructure: APIs & IAM

Prerequisites for Running Class Labs: Before launching Dataproc clusters, VM instances, or AI agents, you must enable the requisite Google APIs and assign proper IAM roles:

Core APIs to Enable

- **Compute Engine API:**
``compute.googleapis.com`` - Core VM instance provisioning.
- **Cloud Dataproc API:**
``dataproc.googleapis.com`` - Managed Spark/Hadoop clusters.
- **Resource Manager API:**
``cloudresourcemanager.googleapis.com`` - IAM bindings.
- **Vertex AI API:**
``aiplatform.googleapis.com`` - Gemini models & Antigravity enterprise agent access.

CLI ACTIVATION & IAM BINDING

```
# 1. Enable APIs via gcloud CLI:
gcloud services enable \
  compute.googleapis.com \
  dataproc.googleapis.com \
  cloudresourcemanager.googleapis.com \
  aiplatform.googleapis.com

# 2. Assign Required IAM Roles in IAM Admin:
# Ensure your user/service account holds:
# - Owner (for project management)
# - Storage Admin (for full GCS read/write access)

gcloud projects add-iam-policy-binding YOUR_PROJECT \
  --member="user:YOUR_GMAIL@gmail.com" \
  --role="roles/storage.admin"
```

Developer and Management Tools Overview

Four Primary Ways to Interact With Google Cloud: Depending on your task (exploratory prototyping, automated scripting, or production deployment), GCP offers four distinct interaction surfaces:

1. Cloud Console (Web UI)

- **Usage:** Interactive visual management, billing setup, and real-time dashboard monitoring.
- **Best For:** Initial exploration, debugging, and visual inspections.

2. Google Cloud Shell

- **Usage:** In-browser Linux terminal pre-authenticated with your account and pre-loaded with developer tools.
- **Best For:** Quick admin tasks without installing local software.

3. Google Cloud CLI (`gcloud`)

- **Usage:** Native command-line interface installed on your personal macOS, Linux, or Windows machine.
- **Best For:** Automation, scripting, local development, and reproducible workflows.

4. Client Libraries & SDKs

- **Usage:** Idiomatic client libraries in Python, Go, Node.js, and Java calling REST/gRPC endpoints.
- **Best For:** Programmatic app development and LLM agent integration.

Official Video Guide: <https://www.youtube.com/watch?v=3v9XWdFyh2E>

Google Cloud Shell Deep Dive

What is Cloud Shell? A complimentary, Debian-based virtual machine provisioned on-demand by Google with 5 GB of persistent home directory storage.

Pre-Installed Tooling: Comes pre-configured with ``gcloud``, ``docker``, ``git``, ``python3``, ``pip``, ``terraform``, ``kubectl``, and code editors.

Critical Operational Rules & Limitations to Remember:

Cloud Shell Inactivity & Persistence Constraints

- **60-Minute Inactivity Timeout:** A Cloud Shell session automatically terminates after 1 hour of terminal inactivity.
- **Non-Home Directory Wipe:** Any modifications made outside your ``$HOME`` directory (e.g. software installed in ``/usr/local`` or ``/tmp``) are completely lost when the instance terminates.
- **5 GB Persistent ``$HOME``:** Only files stored inside your ``/home/<user>`` directory are preserved between sessions.
- **120-Day Dormancy Rule:** If you do not launch Cloud Shell for 120 days, your persistent home disk will be automatically deleted.

Modern GCP Tooling vs. Deprecated Infrastructure

[WARNING] Google Infrastructure Modernization (Fall 2026)

Google has updated core developer tooling and shut down older services found in legacy recitations. Use this transition guide to ensure your code and commands remain fully functional.

Legacy / Deprecated Tools

- **Cloud Datalab:** Legacy containerized Jupyter environment (officially deprecated).
- **Cloud ML Engine:** ``gcloud ml-engine`` command tree discontinued.
- **Cloud IoT Core:** Completely shut down by Google in August 2023.
- **Legacy ``gsutil`` CLI:** Python-based single-threaded storage tool.
- **Cloud Functions (Gen 1):** Legacy event-driven compute engine.

Modern 2026 Replacement Standard

- **Vertex AI Workbench:** Enterprise managed JupyterLab instances with full Git and GPU support.
- **Vertex AI Platform:** Unified ``gcloud ai`` suite for training, fine-tuning, and Gemini models.
- **Discontinued:** Removed from modern curricula; use `pub/sub` + Cloud Run.
- **Modern ``gcloud storage``:** Multi-threaded C++ engine offering up to 94x faster throughput.
- **Cloud Run Functions:** Modern serverless containers running on Cloud Run infrastructure.

Installing the Google Cloud CLI (`gcloud`)

Local CLI Installation: Installing `gcloud` on your local laptop allows you to script deployments, interact with Cloud Storage, and authenticate local AI agents directly.

Platform Installation Instructions:

MACOS / LINUX INSTALLATION

```
# Option A: Homebrew (macOS)
brew install --cask google-cloud-sdk

# Option B: Interactive Curl (macOS/Linux)
curl https://sdk.cloud.google.com | bash
exec -l $SHELL

# Verify installation:
gcloud version
```

WINDOWS INSTALLATION

```
# Windows PowerShell (Run as Administrator):
(New-Object Net.WebClient).DownloadFile(
    "https://dl.google.com/dl/cloudsdk/channels/rapid/
    GoogleCloudSDKInstaller.exe",
    "$env:Temp\GoogleCloudSDKInstaller.exe"
)
& $env:Temp\GoogleCloudSDKInstaller.exe

# Or standalone interactive installer:
# https://cloud.google.com/sdk/docs/install
```

Official Video Guide: <https://docs.cloud.google.com/sdk/docs/install-sdk>

Initializing and Configuring `gcloud`

The `gcloud init` Workflow: The primary setup command that authorizes Cloud SDK tools and establishes your active working configuration.

What `gcloud init` Configures

- **Web Authentication:** Opens your local default browser to log in with your course personal Gmail account.
- **Project Selection:** Lists accessible projects and binds your active CLI session to your course Project ID.
- **Default Compute Region:** Prompts for your default geographical region (select `us-east1` or `us-central1`).
- **Default Compute Zone:** Sets default zone (e.g. `us-east1-b`) so you don't need to specify `--zone` on every command.

ESSENTIAL CONFIGURATION COMMANDS

- # 1. Initialize configuration:
`gcloud init`
- # 2. View active settings:
`gcloud config list`
- # 3. Switch project manually if needed:
`gcloud config set project YOUR_PROJECT_ID`
- # 4. View authenticated accounts:
`gcloud auth list`
- # 5. List and install optional CLI components:
`gcloud components list`
`gcloud components install alpha beta`

Essential `gcloud` Commands & Verification

Core CLI Management Commands

- **Account Login:** `gcloud auth login` - Authorizes the CLI using your personal Gmail account via browser.
- **Project Binding:** `gcloud config set project <PROJECT\_ID>` - Switches active course project context.
- **Compute Region:** `gcloud config set compute/region us-east1` - Sets default geographical region.
- **Active Settings:** `gcloud config list` - Displays active account, project, and region settings.
- **List Projects:** `gcloud projects list` - Verifies all GCP projects accessible by your account.

When is ADC (`application-default login`) Used?

- **What is ADC?:** Application Default Credentials (ADC) is used specifically by programmatic code libraries.
- **Python Client Libraries:** Required when writing custom Python scripts importing `google.cloud.storage` or custom SDKs.
- **How to Generate:** Run `gcloud auth application-default login` to cache local JSON credentials for code libraries.
- **Not Required for `agy`:** Note: The Antigravity CLI uses its own native browser OAuth login and does NOT require ADC setup!
- **Security Note:** ADC securely replaces downloading raw JSON service account keys to your local machine.

Platform Overview: Compute Landscape

GCP Compute Spectrum: Google Cloud provides compute options spanning from full hardware control to fully managed serverless runtimes:

Compute Engine (IaaS)

- **Control:** Full root access to Linux/Windows virtual machines.
- **Customization:** Configurable CPU, RAM, GPU accelerators, and persistent disks.

Dataproc Clusters

- **Control:** Managed cluster for Apache Spark, Hadoop, Flink, and Jupyter.
- **Speed:** Spins up multi-node distributed data environments in under 90 seconds.

Google Kubernetes Engine (GKE)

- **Control:** Production container orchestration using managed Kubernetes.
- **Scale:** Autoscaling microservices and large-scale multi-GPU training jobs.

Cloud Run (Serverless Containers)

- **Control:** Deploy any containerized web app or GenAI API wrapper directly.
- **Economics:** Scales to zero when idle - costs \$0 when not processing requests.

Official Video Guide: <https://www.youtube.com/watch?v=hHmBOv8z1D0>

Compute Engine: Virtual Machines & Families

Compute Engine Instances: Secure, customizable, software-based computers running on Google's physical server infrastructure. You select: Operating System, CPU, RAM, Disk, and Region.

Google Cloud Machine Families (2026):

General Purpose (E2, N2, N4)

- **E2 Series:** Cost-effective, shared-core or standard instances. Default for coursework labs (e.g. `e2-standard-4`).
- **N2 / N4 Series:** Balanced performance, flexible vCPU/RAM ratios for production microservices.

Compute-Optimized (C3, C4)

- **Architecture:** Powered by Intel Xeon / AMD EPYC with Google Titanium IPU.
- **Use Cases:** High-performance computing (HPC), gaming servers, and intensive batch processing.

Memory-Optimized (M3)

- **Architecture:** Ultra-high memory capacity (up to multiple terabytes of RAM).
- **Use Cases:** Large in-memory databases (SAP HANA) and giant caching clusters.

Accelerator-Optimized (A2, A3, G2)

- **G2 Series:** NVIDIA L4 GPUs for cost-effective GenAI inference and graphics.
- **A2 / A3 Series:** NVIDIA A100 / H100 GPU clusters designed for LLM training.

VM Storage & Virtual Private Cloud (VPC)

Compute Engine Storage Options

- **Persistent Disks (Standard):** ``pd-standard`` - Magnetic HDD storage. Economical for large sequential datasets and backups.
- **Balanced Persistent Disks:** ``pd-balanced`` - Default SSD choice. Reliable IOPS for boot disks and general development.
- **SSD Persistent Disks:** ``pd-ssd`` - High-performance IOPS for enterprise relational databases.
- **Local SSD:** Physically attached NVMe drive. Ultra-low latency, but ephemeral (data is erased when the VM stops).
- **Cloud Storage Buckets:** Object storage accessible over HTTP from any VM, container, or local machine.

Virtual Private Cloud (VPC) Networking

- **Internal IP Addresses:** Private IP assigned to the VM for internal VPC communication (free network egress within same zone).
- **External IP Addresses:** Publicly routable IP required to SSH from the open internet or serve public web traffic.
- **VPC Networks & Subnets:** Isolated virtual network defining IP ranges across global GCP regions.
- **Firewall Rules:** Stateful security rules governing ingress/egress traffic by port and protocol (e.g. allow port 22 for SSH, port 8888 for Jupyter).
- **Cloud Load Balancing:** Distributes incoming traffic across multiple VM instances for high availability.

Dataprox Managed Clusters for Spark & AI

Single-Node Academic Configuration

- **Managed Spark & Hadoop:** Automated deployment, scaling, and lifecycle management for big data jobs.
- **Single-Node Topology:** Runs master and worker daemons on one instance () to minimize credit burn.
- **Machine Sizing:** (4 vCPUs, 16 GB RAM) with 100 GB standard persistent disk.
- **Modern Debian 13 Image:** Uses with up-to-date Python 3 and Spark runtimes.
- **Component Gateway:** Provides zero-configuration secure web access to JupyterLab and Spark Web UI without SSH tunnels!

FALL 2026 PRODUCTION CLI COMMAND

```
# Create Single-Node Dataprox Cluster:
gcloud dataproc clusters create cluster-rec1 --
project=YOUR_PROJECT_ID --region=us-east1 --single-node
--public-ip-address --master-machine-type=e2-standard-4 --
master-boot-disk-type=pd-standard --master-boot-disk-
size=100GB --image-version=3.0-debian13 --optional-
components=DOCKER,JUPYTER --enable-component-gateway
--scopes=https://www.googleapis.com/auth/cloud-platform

# Open JupyterLab in Cloud Console:
# Dataprox > Clusters > cluster-rec1 >
# 'Web Interfaces' > JupyterLab
```

Cloud Run: Modern Serverless Execution

Evolution of Serverless at Google: Cloud Functions has evolved into Cloud Run functions. Today, Cloud Run is Google's flagship container execution platform.

Key Advantages for GenAI Applications:

Why Developers Prefer Cloud Run

- **Any Programming Language:** Package any Python (FastAPI/Flask), Node.js, or Go service inside a standard container.
- **True Scale-to-Zero:** When no requests arrive, Cloud Run scales down to 0 instances, incurring zero cost while idle.
- **Automatic HTTPS & TLS:** Google provisions a globally routed HTTPS endpoint with valid SSL certificates out-of-the-box.
- **Concurrency Support:** A single container instance can handle up to 250 simultaneous requests.

ONE-COMMAND DEPLOYMENT

```
# Deploy a GenAI container service to Cloud Run:
gcloud run deploy gemini-agent-service \
  --source . \
  --region us-east1 \
  --allow-unauthenticated \
  --set-env-vars GOOGLE_GENAI_USE_VERTEXAI=true

# Cloud Run automatically builds the container,
# deploys to Google's serverless fleet, and returns
# a live public HTTPS URL in seconds.
```

Google Cloud Storage: Object Storage

Object Storage Fundamentals: Google Cloud Storage (GCS) stores unstructured data as objects inside containers called buckets (conceptually like an infinite 'Google Drive for cloud workloads').

Core Storage Characteristics:

Bucket Architecture & Naming

- **Global Unique Namespace:** Bucket names must be globally unique across ALL Google Cloud customers worldwide (e.g. `cmu-genai-f26-student`).
- **Objects & Immutability:** Files stored as objects. When an object is modified, the entire object is overwritten.
- **High Durability (11 9's):** Designed for 99.999999999% annual durability through multi-datacenter replication.
- **Fine-Grained Access Control:** Enforce permissions using IAM policies and signed URLs for temporary client downloads.

4 Storage Tiers by Access Frequency

- **Standard Storage:** Best for active data accessed frequently (datasets, model weights, web assets).
- **Nearline Storage:** Data accessed at most once every 30 days. Lower storage cost, small retrieval fee.
- **Coldline Storage:** Data accessed at most once every 90 days. Ideal for quarterly disaster recovery.
- **Archive Storage:** Data accessed less than once a year. Lowest storage cost for permanent archives.

Official Video Guide: <https://www.youtube.com/watch?v=tc2940Zwvyk>

Modern Storage CLI: `gcloud storage`

[IMPORTANT] Upgrade from Legacy `gsutil` to `gcloud storage`

`gcloud storage` is Google's high-throughput CLI tool. It utilizes a fast parallel transfer engine that delivers up to 94% faster on downloads and 57% faster on uploads (Google's published benchmark, single 10GB file) compared to legacy `gsutil`.

ESSENTIAL `GLOUD STORAGE` COMMANDS

1. Create a new Cloud Storage bucket in us-east1:

```
gcloud storage buckets create gs://cmu-genai-f26-shashank --location=us-east1
```

2. Upload a file or an entire directory in parallel:

```
gcloud storage cp -r ./training_data gs://cmu-genai-f26-shashank/
```

3. List bucket contents:

```
gcloud storage ls --recursive gs://cmu-genai-f26-shashank/
```

4. Download files locally:

```
gcloud storage cp gs://cmu-genai-f26-shashank/training_data/dataset.json ./local_data/
```

5. Delete an object or remove an entire bucket:

```
gcloud storage rm -r gs://cmu-genai-f26-shashank/
```

Databases & Analytical Warehousing

Google Cloud Managed Data Services: Depending on query complexity, scale, and transaction models, GCP offers specialized database engines:

BigQuery (Analytics)

- **Model:** Serverless, petabyte-scale enterprise SQL data warehouse.
- **GenAI:** Native BigQuery ML with integrated Gemini vector embeddings and search.

Cloud SQL & AlloyDB (Relational)

- **Model:** Fully managed PostgreSQL, MySQL, and SQL Server.
- **AlloyDB:** Google's ultra-high-performance PostgreSQL engine for heavy workloads.

Cloud Spanner (Global Relational)

- **Model:** Globally distributed database offering full ACID relational consistency.
- **Availability:** Industry-leading 99.999% availability SLA with seamless horizontal scale.

Firestore (NoSQL Document Store)

- **Model:** Serverless NoSQL document database with real-time data sync.
- **Use Cases:** Fast state management for mobile apps, user sessions, and web clients.

How to Use Google Cloud Console (Lab Walkthrough)

Step-by-Step Hands-On Demonstration

Step 1: Project Verification: Open the Console and confirm that your designated course project is selected in the top navigation bar.

Step 2: GCS Bucket Creation: Navigate to Cloud Storage > Buckets > Create. Choose a unique name, select region `us-east1`, Standard storage, and Uniform access control.

Step 3: Staging Lab Data: Drag and drop a test file or dataset into the newly created bucket. Observe access permissions and object metadata.

Step 4: Compute Engine Provisioning: Navigate to Compute Engine > VM Instances > Create Instance. Select `e2-standard-4`, Debian 13 OS boot disk, and check 'Allow HTTP/HTTPS traffic'.

Step 5: Browser-Based SSH Connection: Click the 'SSH' button directly in the VM instances table to open an encrypted terminal in a pop-up window.

Step 6: Proper Resource Tear-Down: Stop the VM instance immediately upon completing the verification to ensure 0 lingering charges.

Google Cloud IAM Architecture

Identity and Access Management (IAM): IAM authorizes WHO can take WHAT action on WHICH resource across your cloud environment.

The Three Core Pillars of IAM:

1. WHO (Principal)

- **Google Account:** Individual user email (e.g. `student@gmail.com`).
- **Google Group:** Collection of Google accounts (e.g. `cmu-tas@cmu.edu`).
- **Service Account:** Machine identity used by applications and agents.
- **Workspace Domain:** All users within an organization.

2. WHAT (Role)

- **Basic Roles:** Owner, Editor, Viewer. (Broad, legacy, avoid in production).
- **Predefined Roles:** Granular Google-managed roles (e.g. `Storage Object Admin`).
- **Custom Roles:** User-tailored sets of exact individual permissions.

3. ON WHAT (Resource)

- **Organization:** Top-level root node representing the university/enterprise.
- **Folders:** Departmental or course groupings.
- **Projects:** The resource container.
- **Inheritance:** Permissions inherit downwards from Org to Project!

Official Video Guide: <https://www.youtube.com/watch?v=PqMGmRhKsnM>

IAM Security Best Practices

Principle of Least Privilege: Always grant only the minimum permissions necessary to perform a task. Never grant broad administrative access when specific roles suffice.

Avoid Assigning Basic Owner / Editor Roles: Do not assign the `Owner` role to classmates or applications. Instead, assign granular predefined roles (e.g. `Storage Object Viewer` or `Dataproc Worker`).

Separate Environments by Project: Maintain strict separation between experimentation projects and production/evaluation projects to prevent accidental data overwrites.

Audit Regularly with IAM Recommender: Google Cloud's built-in IAM Recommender automatically analyzes access patterns and suggests pruning unused permissions.

gcloud components list

Your current Cloud SDK version is: 222.0.0
The latest available version is: 222.0.0

Components	
Status	Name
Not Installed	Cloud Bigtable Command Line Tool
Not Installed	Cloud Bigtable Emulator
Not Installed	Cloud Datalab Command Line Tool
Not Installed	Cloud Pub/Sub Emulator
Not Installed	Cloud SQL Proxy
Installed	App Engine Go Extensions
Installed	gcloud app Java Extensions
Installed	gcloud app Python Extensions
Installed	kubectl

Service Accounts & Secret Management

What is a Service Account? A special Google Cloud identity used by non-human callers, such as virtual machines, automated CI/CD pipelines, and autonomous AI agents to interact with APIs.

Identified by Email Address: `[account-name]@[project-id].iam.gserviceaccount.com`

Security Warning on JSON Keys:

[WARNING] CRITICAL SECURITY ALERT: NEVER Commit Service Account Keys to Git

Automated bots continuously scrape public GitHub repositories. Leaking a service account key can allow malicious actors to compromise your cloud account, spin up illicit crypto-mining clusters, and consume thousands of dollars in minutes.

Safe Modern Alternative: Always use Application Default Credentials (ADC) via 'gcloud auth application-default login' on your local development machine.

Google Cloud Logging (Observability)

Centralized Telemetry Aggregation: Google Cloud Logging automatically collects, stores, and indexes logs from every service across your cloud projects in real time.

Why Cloud Logs are Vital:

Rapid Debugging: Instantly inspect why a Python script crashed, why a Cloud Run container failed to start, or why a Dataproc job was rejected.

Performance Monitoring: Identify memory bottlenecks and trace slow API responses.

Security & Compliance Auditing: Audit logs track every administrative action: who modified firewall rules, who downloaded a bucket object, and when.

Primary Ingestion Sources: Compute Engine VMs, GCS Buckets, Cloud Run, Dataproc Spark jobs, and IAM authorization services.

Logs Explorer & Diagnostic Querying

Logs Explorer Query Concepts

- **Navigation:** In Cloud Console: Navigation Menu > View all products > Observability > Logging > Logs Explorer
- **Resource Filtering:** Pinpoint logs by resource type (e.g. gce_instance, gcs_bucket, cloud_run_revision).
- **Severity Hierarchy:** DEFAULT (0) < DEBUG (100) < INFO (200) < NOTICE (300) < WARNING (400) < ERROR (500) < CRITICAL (600) < ALERT (700) < EMERGENCY (800)
- **Time Bounding:** Always bound queries to narrow intervals (last 15m, 1h) to speed up index retrieval.
- **Free-Text Search:** Use regex matching (=~) to locate exact exception names or crash dumps.

PRODUCTION DIAGNOSTIC QUERIES

```
# Example 1: Compute Engine VM Errors
resource.type="gce_instance"
severity>=ERROR
timestamp >= "2026-09-01T00:00:00Z"
```

```
# Example 2: Cloud Storage Access Failures
resource.type="gcs_bucket"
protoPayload.status.code=7
# (Code 7 = PERMISSION_DENIED)
```

```
# Example 3: Cloud Run Application Crashes
resource.type="cloud_run_revision"
severity>=WARNING
textPayload =~ "(?!exception|timeout)"
```

```
# Example 4: Dataproc Spark Job Failures
resource.type="cloud_dataproc_cluster"
severity=ERROR
```

5-Step Cloud Troubleshooting Workflow

Systematic Diagnostic Procedure for Cloud Failures

Step 1: Read the Exact Error: Carefully read the error banner in the Console UI or the exit message in your terminal. Note the error code (e.g. 403 Forbidden, 404 Not Found, 503 Unavailable).

Step 2: Open Logs Explorer: Navigate to Logging > Logs Explorer. Select your active course Project ID and narrow the time range to the last 15–30 minutes.

Step 3: Filter by Resource & Severity: Set the filter to ``severity >= ERROR`` and select the affected resource type (e.g. ``gce_instance``, ``cloud_dataproc_cluster``).

Step 4: Trace the Earliest Error Message: In distributed cloud systems, a single failure cascades into dozens of downstream errors. Always sort by timestamp ascending and identify the FIRST error!

Step 5: Remediate the Root Cause: Most student issues fall into three categories: (1) Missing/disabled API, (2) Insufficient IAM role, or (3) Suspended billing account.

Introduction to Antigravity CLI (`agy`)

What is Google Antigravity? An agent-first software engineering platform engineered by Google DeepMind and Cloud teams for autonomous planning, multi-file code execution, test verification, and browser automation.

The Antigravity CLI (`agy`): An ultra-fast, terminal-native agent harness for fast developer workflows, scripting, and CI/CD pipelines.

Cross-Platform Installation Scripts:

MACOS / LINUX INSTALLATION

```
# macOS & Linux Native Installer:
curl -fsSL https://antigravity.google/cli/install.sh | bash

# Installs directly to:
# ~/.local/bin/agy
# And automatically configures shell profile (~/.zshrc)

# Verify installation:
agy --version
```

WINDOWS INSTALLATION

```
# Windows PowerShell (Run as Administrator):
irm https://antigravity.google/cli/install.ps1 | iex

# Installs to:
# C:\Users\<username>\AppData\Local\agy\bin

# Command Prompt (CMD):
curl -fsSL https://antigravity.google/cli/install.cmd -o install.cmd &&
install.cmd && del install.cmd
```

Official Video Guide: <https://antigravity.google/download#antigravity-cli>

Integrating Antigravity with VS Code

VS Code Extension Setup

- 1. Open VS Code:** Launch Visual Studio Code (version 1.90 or later).
- 2. Open Extensions View:** Press ``Cmd+Shift+X`` (macOS) or ``Ctrl+Shift+X`` (Windows/Linux).
- 3. Search 'Google Antigravity':** Locate the official extension published by Google.
- 4. Click Install:** The extension automatically bundles the local ``agy`` background service.
- 5. Open Agent Side Panel:** Click the Antigravity icon in the left Activity Bar to launch.

Key Features Inside VS Code

- **Conversational Agent Panel:** Chat with Gemini models directly alongside your open code files.
- **Granular Inline Diffs:** Inspect color-coded diffs before accepting agent code proposals.
- **Review Changes Editor:** Review and approve multi-file modifications in a unified view.
- **Shared CLI & IDE Harness:** Shares persistent memory, skills, and configuration with the ``agy`` CLI.

Official Video Guide: <https://antigravity.google/docs/ide/extensions/vscode/>

Essential Antigravity CLI Commands & Shortcuts

CLI Launch & Terminal Shortcuts

- **Interactive Session:** Run ``agy`` in your project folder to enter the full-screen terminal UI.
- **Headless One-Shot Mode:** Run `agy '<prompt>'` to execute a task and automatically exit upon completion.
- **Shell Execution (!):** Prefix with `!`` to execute terminal commands directly (e.g. `!git status``, `!ls -la``).
- **File Autocomplete (@):** Type ``@`` to trigger intelligent workspace file path suggestions.
- **Clear Input Prompt:** Press ``esc` + `esc`` to clear the current prompt input box.
- **Exit CLI Session:** Press ``Ctrl+D`` or type ``/exit`` or ``/quit`` to terminate.

Core Slash Commands (Type ``?`` or ``/ help``)

- **``/help`` or ``?``:** Display comprehensive list of all active slash commands and keybindings.
- **``/config`` (or ``/settings``):** Open full-screen settings menu to configure workspace behavior and models.
- **``/clear``:** Clear conversation history and reset prompt context for a fresh task.
- **``/rewind`` (or ``/undo``):** Roll back conversation history to a previous step or state.
- **``/fork``:** Branch current conversation into a separate isolated workspace session.
- **``/resume``:** List and restore prior conversation sessions and logs.
- **``/model``:** Switch the active AI foundation model during an active session.
- **``/logout``:** Disconnect Google Cloud session, clear cached tokens, and reset authentication.

Development Modes & Milestone Building

1. Autopilot Mode

- **Definition:** Agent-driven autonomy.
- **Action:** The AI writes code, creates files, and runs tests automatically based on instructions.

2. Review-Driven

- **Definition:** Full human-in-the-loop oversight.
- **Action:** AI requests explicit user permission before executing terminal commands or edits.

3. Agent-Assisted

- **Definition:** Collaborative pair programming.
- **Action:** You write code directly; the AI manages boilerplate, unit tests, and safe automations.

[IMPORTANT] Best Practice: Milestone-Based Iterative Building to Avoid Rate Limits

Antigravity enforces rate limits to ensure fair resource allocation. Never ask the AI to build an entire application in a single monolithic prompt. Instead, break your specification into 3 distinct milestones (e.g. Milestone 1: Data Models, Milestone 2: Core Logic, Milestone 3: API/UI). Build and verify each milestone incrementally.

Connecting to Vertex AI (NOT AI Studio)

[WARNING] CRITICAL POLICY REQUIREMENT: Vertex AI (Gemini Enterprise) ONLY

Students must connect Antigravity CLI to Google Cloud Vertex AI using course GCP Project credentials. Do NOT use personal Google AI Studio API keys for course assignments.

Google AI Studio Keys (Forbidden)

- **Personal Billing:** Billed to personal credit cards; cannot draw from course educational coupons!
- **Strict Rate Limits:** Heavily throttled per-minute request ceilings unsuitable for agentic workflows.
- **No Data Governance:** Hobbyist tier without enterprise confidentiality guarantees.
- **Unmanaged:** Cannot be governed by course IAM policies or audit logging.

Vertex AI / Gemini Enterprise (Required)

- **Course Coupon Funded:** Directly billed to your course GCP Project ID, drawing from educational grant credits.
- **Enterprise SLA & Quota:** High-throughput token limits designed for multi-agent loops.
- **Data Privacy Guarantee:** Customer data and code prompts are NEVER used to train foundation models.
- **Unified IAM & Security:** Secured via Google Cloud IAM and project policies.

`agy` Interactive Authentication Walkthrough

[INFO] Zero Environment Variables Required

Antigravity CLI authenticates directly via interactive browser OAuth. You do NOT need to set environment variables (GOOGLE_GENAI_USE_VERTEXAI, etc.) or manually edit settings files.

Step-by-Step Authentication Flow

- 1. Launch the CLI:** In your project directory, open your terminal and simply run: ``agy``
- 2. Select Google Cloud:** On the welcome screen, highlight 'Continue with Google Cloud' and press Enter.
- 3. Browser Sign-In:** Your default browser opens automatically. Sign in with your course personal Gmail account.
- 4. Select Project & Region:** In the browser prompt, select your course Project ID (funded by coupons) and default region (``us-central1``).

Verification & Session Management

- **Verify in Header:** Upon returning to terminal, the ``agy`` top status bar confirms your active Project ID and Vertex AI connection.
- **Persistent Token Caching:** Credentials are secure and cached locally. Subsequent ``agy`` runs start instantly without re-prompting.
- **Switching Projects / Accounts:** If you need to change project or re-authenticate, simply run ``/logout`` inside ``agy``.
- **VS Code Extension:** The official VS Code extension automatically inherits the authenticated session from the CLI!

Installing Agent Skills in Antigravity

- An Agent Skill is a folder containing a SKILL.md file - instructions that teach Antigravity how to handle a specific type of task
- Antigravity auto-discovers skills placed in `.agents/skills/<skill-name>/` - no registration or config needed
- Skills repo for this course: github.com/Systems-and-Toolchains-for-AI-Engineers/agentic-skills

INSTALLING A SKILL

```
# Clone Repository
git clone https://github.com/Systems-and-Toolchains-for-AI-Engineers/agentic-skills

# Create Skills Directory
mkdir -p .agents/skills
cp -R agentic-skills/database-reviewer .agents/skills/

# Start Antigravity
agy

# Confirm Skill Loaded with Example Prompt
review this SQL query for issues: SELECT * FROM users WHERE email = " + userInput + "
```

```
# Home/Downloads/skill-demo/.agents/skills/database-reviewer/SKILL.md (ctrl-c to expand)

• Thought For $s, $m tokens
Revising Security First
New Findings

#### [Blinking] SQL Injection Vulnerability
sqlinject is concatenated directly into the SQL string. Any untreated value can break out of the string literal and execute arbitrary SQL. (e.g., ' OR '1'='1 to bypass checks or UNION SELECT ... to exfiltrate data).

Fix: Use parameterized queries / prepared statements with placeholder binding instead of string concatenation.

- SELECT * FROM users WHERE email = " + userInput + "
+ SELECT * FROM users WHERE email = ?

Example fix language / delivery

• Node.js (e.g. / PostgresDB.js):
const result = await db.query(
  `SELECT id, name, email FROM users WHERE email = ${}`,
  [userInput]
);

• Python (example):
def get_users(email):
    """SELECT id, name, email FROM users WHERE email = %s"""
    return db.query(email)

#### [Should Fix] Overfetching with SELECT
Using SELECT * retrieves every column in the users table:

• Defaults covering indexes (index-only scans).
• Unnecessarily exposes sensitive user fields (e.g., password_hash, reset_token, MFA secrets).
• Increases network payload and memory overhead.

Fix: Explicitly project only the specific columns your application needs:
SELECT id, name, email FROM users WHERE email = ?

#### [Should Fix] Indexing and Case Sensitivity on email

1. Index Coverage: Ensure email has a unique index to avoid a full table scan (log below):
CREATE INDEX idx_email ON users(email);

2. Case Normalization: If email matching should be case-insensitive, avoid wrapping the column in a function in queries like WHERE LOWER(email) = LOWER(?) unless backed by an expression index:
SELECT email FROM users WHERE LOWER(email) = LOWER(?);
Alternatively, use the ILIKE extension for the user column.
```

Recitation Summary & Student Checklist

Student Action Checklist for Next Lecture

- **Personal Gmail Ready:** Confirm access to personal Gmail without active \$300 trial conflicts.
- **Watch for Coupon Email:** Coupons will be distributed in the upcoming lecture. Verify Andrew email link immediately.
- **Redeem & Validate:** Redeem coupon at console.cloud.google.com/education under personal Gmail.
- **Configure \$30 Budget Alert:** Set up \$30 budget with 50%, 90%, and 100% email threshold alerts.
- **Install Google Cloud CLI:** Install `gcloud` and run `gcloud init` to configure your default project.
- **Install Antigravity CLI:** Install `agy` via native installer and install official VS Code extension.
- **Authenticate with Google Cloud:** Run `agy` -> Select 'Continue with Google Cloud' -> link course Project ID in browser.

Support Channels & Resources

- **Office Hours:** Weekly TA office hours posted on Canvas.
- **Discussion Board:** Post technical questions on Piazza / Canvas Discussions.
- **Credit Issues:** Email Prof. Mohamed Farag at farag@cmu.edu (allow 24-48h lead time).
- **Google Cloud Docs:** <https://cloud.google.com/docs>
- **Antigravity Documentation:** <https://antigravity.google/docs/cli/overview/>
- **Course Staff:** TA Shashank Aluru, Serena Gomez | Prof. Mohamed Farag